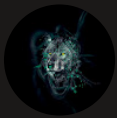




Creating a Private Subnet

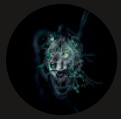


Kiprono Yegon

The screenshot shows the 'Edit subnet settings' page in the AWS Management Console. The page is titled 'Edit subnet settings' and includes a breadcrumb trail 'Subnets > Edit subnet settings'. The main content area is divided into several sections:

- Subnet:** Displays the Subnet ID 'subnet-032c3dae1f3be973d' and the Name 'privatesub'.
- Auto-assign IP settings:** Includes a description 'Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.' and two checkboxes: 'Enable auto-assign public IPv4 address' (unchecked) and 'Enable auto-assign customer-owned IPv4 address' (checked, with a note 'Option disabled because no customer owned pools found.').
- Resource-based name (RBN) settings:** Includes a description 'Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.' and two checkboxes: 'Enable resource name DNS A record on launch' (unchecked) and 'Enable resource name DNS AAAA record on launch' (checked). Below these is a 'Hostname type' section with two radio buttons: 'Resource name' (unchecked) and 'IP name' (checked).
- DNS64 settings:** Includes a description 'Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.' and a checkbox 'Enable DNS64' (unchecked).

At the bottom right of the form, there are 'Cancel' and 'Save' buttons.



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a cloud service by Amazon that allows you to create a private, isolated network within the cloud where you can securely run your applications and resources. It is useful because it gives you full control over your networking environment, including IP address ranges, subnets, routing, and security settings, enabling you to protect your systems, manage traffic flow, and design scalable architectures.

How I used Amazon VPC in this project

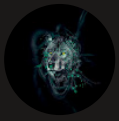
In today's project, I used Amazon VPC to create a private subnet that does not connect with the internet via the internet gateway

One thing I didn't expect in this project was...

One thing I didn't expect in this project is the creation of private subnets and private route tables to supplement the private subnet

This project took me...

This project took me 2 hours since I had to start all over



Private vs Public Subnets

The difference between public and private subnets is that resources in the public subnet are accessed over the internet whereas the resources in the private subnet are only accessible within the vpc

Having private subnets are useful because they are used to keep resources that are private within a VPC such as the database

My private and public subnets cannot have the same IP address because IP addresses must be unique within a VPC and it prevents CIDR blocks do not overlap

The screenshot shows the 'Edit subnet settings' page in the AWS Management Console. The page is titled 'Edit subnet settings' and includes a breadcrumb trail 'v273d > Edit subnet settings'. The main content area is divided into four sections: 'Subnet', 'Auto-assign IP settings', 'Resource-based name (RBN) settings', and 'DNS64 settings'. The 'Subnet' section shows the Subnet ID 'subnet-032c3de1f73be973d' and the Name 'privatesub'. The 'Auto-assign IP settings' section has a description 'Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an Instance in this subnet.' and two options: 'Enable auto-assign public IPv4 address' (unchecked) and 'Enable auto-assign customer-owned IPv4 address' (checked). The 'Resource-based name (RBN) settings' section has a description 'Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.' and two options: 'Enable resource name DNS A record on launch' (unchecked) and 'Enable resource name DNS AAAA record on launch' (checked). The 'DNS64 settings' section has a description 'Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.' and one option: 'Enable DNS64' (checked). At the bottom right, there are 'Cancel' and 'Save' buttons.

Subnet settings

Subnet ID: subnet-032c3de1f73be973d

Name: privatesub

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an Instance in this subnet.

☐ Enable auto-assign public IPv4 address

☒ Enable auto-assign customer-owned IPv4 address

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch

☒ Enable resource name DNS AAAA record on launch

Hostname type

☐ Resource name

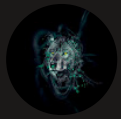
☒ IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☒ Enable DNS64

Cancel Save



A dedicated route table

By default, my private subnet is associated with the VPC's main route table

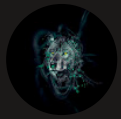
I had to set up a new route table because the private subnet should not access the Internet Gateway

My private subnet's dedicated route table only has one inbound and one outbound rule that allows traffic only from within the VPC, no internet inbound

The screenshot shows the AWS Management Console interface. At the top, there's a section for 'Route tables (1/4)' with a search bar and a table listing route tables. The table has columns: Name, Route table ID, Explicit subnet associations, Edge associations, Main, VPC, and Owner ID. The 'danPrivate-rtb' route table is selected. Below this, there's a section for 'rtb-0a1a02b05fb99d1ef / danPrivate-rtb' with tabs for Details, Routes, Subnet associations, Edge associations, Route propagation, and Tags. The 'Subnet associations' tab is active, showing a table for 'Explicit subnet associations (1)'. This table has columns: Name, Subnet ID, IPv4 CIDR, and IPv6 CIDR. The 'privatesub' is associated with 'subnet-032c3dae1f3be973d' and has an IPv4 CIDR of '10.0.1.0/24'. Below this, there's a section for 'Subnets without explicit associations (0)'.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0c58021a6018225ca	-	-	Yes	vpc-09f97046c805a557	783259868137
-	rtb-0c0c262a2d3109ae4	-	-	Yes	vpc-0874701773a8b99e57 dan...	783259868137
danrtb	rtb-01a022c0b0911a064	subnet-079f5750481501...	-	No	vpc-0874701773a8b99e57 dan...	783259868137
danPrivate-rtb	rtb-0a1a02b05fb99d1ef	subnet-032c3dae1f3be9...	-	No	vpc-0874701773a8b99e57 dan...	783259868137

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
privatesub	subnet-032c3dae1f3be973d	10.0.1.0/24	-

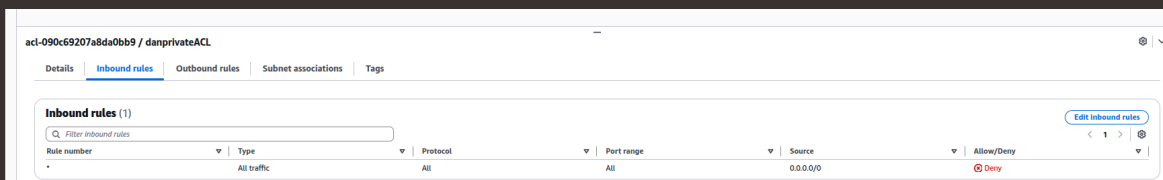


A new network ACL

By default, my private subnet is associated with the default network ACL before setting up an explicit association between the private subnet and another network ACL.

I set up a dedicated network ACL for my private subnet because route table configurations might not be great for security. If any part of the VPC gets compromised the attacker could take advantage of your permissive ACL setup to access or attack the resources in your private subnet!

My new network ACL has two simple rules -no inbound traffic and no outbound traffic





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